

Alexa, what are you not telling me?

There will be an estimated 19.2 billion connected Internet of Things (IoT) devices by the end of 2024, which is predicted to increase year-on-year, with 29.7 billion devices predicted by 2027 (Sinha, 2023). Alongside the growing popularity of IoT devices, there has been a steep rise in the use of Smart Personal Assistants (SPA) to control said devices, particularly since the release of the Amazon Echo in 2014, with sales of Echo smart speakers rising from 0.6 million in 2014 to a predicted 105 million by the end of 2024 (Laricchia, 2022). Amazon has provided a market-leading IoT platform in the way of Amazon Web Services (AWS) IoT, in which Alexa cloud-powered Echo devices control a range of IoT devices in millions of homes around the world (Pathak et al., 2022). While Echo devices allow for the control of IoT devices such as lighting, heating, and televisions, Alexa 'skills' extend this functionality further, allowing for quick and easy access to thousands of third-party apps that can allow users to, for example, play games on their Echo device, quickly find and navigate through recipes, or talk to a virtual personal trainer (Amazon, n.d.). Despite the popularity of this IoT platform for developers and end users alike, many studies have highlighted a plethora of security issues within such IoT platforms, due to their architecture and subsequent overreliance on cloud-based technology (due to the IoT devices' low computational power), which creates a larger attack surface for cybercriminals to exploit (Pathak et al., 2022; Brinson, Winner & Brooker, 2021; Babun et al., 2021). Such security flaws make users vulnerable to a range of attacks which could compromise their data, finances, as well as their connected IoT devices (Brinson, Winner & Brooker, 2021; Pathak et al., 2022). This paper will look at Echo devices, the AWS IoT platform, and Amazon's role within a solution to a third-party skill security vulnerability. This solution involves the user and increases their awareness

of the security risks that they take when they use third-party skills through their device.

The wide-ranging functionality of Echo devices has seen the growth in Echo device sales and, in turn, has attracted third-party developers to increase investment in the development of skills for a growing customer base to access their services, with a total of 199,295 skills from 88,391 developers recorded in 2023 (Edu et al., 2023). To access these skills, users need only tell Alexa to 'open' the required skill from their Echo device (Amazon, n.d.). While Amazon has benefitted financially from opening up their IoT platform to third-party developers, this has given attackers further opportunities to compromise users' devices or personal data (Pathak et al., 2022); 'Skill Squatting Attacks' (Lit, Kim & Si, 2021, p.4) – a method of attack in which malicious code is activated due to it being inadvertently activated with a voice command, due to it having a similar name to a trusted Alexa Skill name - have become increasingly popular amongst cybercriminals (Lit, Kim & Si, 2021; Pathak et al., 2022). This attack method can involve the launching of malicious skills with a phonetically similar name, so as to ensure that the malicious skill is launched as a result of a slight mispronunciation (Pathak et al., 2022; Brinson, Wimmer & Brooker, 2021). Alternatively, the third-party skills can be registered with a skill name that is identical to an existing skill but a 'please' or 'thank you' tagged on the end, so that the malicious skill is launched rather than the intended skill (Pathak et al., 2022; Brinson, Wimmer & Brooker, 2021). In researching further security issues caused by third-party skills, Edu et al. (2023) conducted an analysis of how over 199,000 third-party skills made use of users' data, finding that 43% of the third-party skill developers were not completely transparent about how that data was used.

Additionally, it was discovered that even skills that were deemed to be providing clear traceability of their use of data were able to circumvent Alexa's permission system in order to obtain additional personal information (Edu et al., 2023). In addressing the issue of users activating malicious third-party skills, Edu, Such & Suarez-Tangil (2021) suggest that users should be made more aware of how their information may be shared over SPAs through the publishing of results of privacy assessments undertaken for each third-party app (Edu, Such & Suarez-Tangil, 2021). This study was subsequently followed-up with the development of SkillVet - a program that analyses privacy policies uploaded by the third-party developers and compares these to permissions that are actually requested by the skill through the Alexa API (Edu et al., 2023). The current paper proposes that SkillVet's privacy assessments are combined with additional security ratings calculated by a program – SecureRate – based on how similar a registered skill's name is to an existing skill and its inclusion of key words such as 'please' and 'thank you/thanks'. The SecureRate security rating (given out of 10) would then be made available to the user before they choose to launch that skill, as well as a message which will ask the user to confirm that they would like to launch the named skill.

While Brinson, Wimmer & Brooker (2021) agree that users should be provided with more detailed feedback on their Echo devices - for example to keep users more informed about why their connection to the internet has failed - efficiency and perceived usefulness is one of the strongest factors in a customer's decision to purchase an SPA (Fernandes & Oliveira, 2021); it is therefore important that the suggested changes to the existing processes are implemented as such. The security rating program would therefore need to be fair, robust, yet not a time-consuming

process so as to discourage third-party developers from adding skills. A factor that would lower a security rating would be the tagging of a 'please' or 'thank you' to the end of the registered skill name. Additionally, when registering a new skill name, a search comparison of existing names would be conducted with a rating given based on how similar this name is to an existing name; if a skill name is judged to be over 95% similar to an existing skill name, this could be a strong indication that that skill is trying to achieve an activation phrase that is phonetically similar to an existing skill and thus it would lower the security rating significantly. In addition to this, if each skill is announced to the user before that skill is activated, for example "Are you sure you want to launch the skill 'Uber skill, please?' the user can be alerted to a potentially malicious skill, due to its dubious name choice. If a skill's name is a close phonetic match to an existing skill name, this method could further safeguard the user by making them aware of the recorded mispronunciation, for example, "Are you sure you would like to launch the skill 'Chust Eat?" Combined with the quoted security rating, users would be in a much more empowered position to make choices over which skills to launch, for example, "Are you sure you want to launch the skill 'Chust Eat; Security rating: 0?' (see Figure 1). To further increase efficiency and avoid a situation wherein the user must repeat their skill name several times, Alexa would also suggest the closest match with the highest security rating, for example "Instead, would you like to launch the skill 'Just Eat; security rating 10?' (see Figure 1). Upon implementing the new security-rating system with skills that are already registered, where skills have a similar name, the earlier registered skill would be given the higher security rating. Developers receiving a low security rating for their skill would be prompted to change their name before this rating is formally registered, with the similarity issue communicated, so as not to discourage further skill development.

However, with efficiency mind in terms of companies wanting a particular launching word that will not be overly onerous, the user would be given the opportunity to attach a word or phrase to this skill, for example: “Would you like to save the skill ‘Affordable Taxis’ as the name ‘taxi’?” (developers would be able to provide a suggested alternative name); additionally, there would also be a record of which skills have been previously launched by the user so that the confirmation feature can be bypassed if a skill request matches the user’s previously launched skills.

However, data stored in one location in the cloud can be vulnerable, for example ‘digital forensics’ attacks have seen users’ voice command data manipulated so that they appear to have requested information associated with illegal activity (Brinson, Wimmer & Brooker, 2021). It would therefore be remiss of Amazon not to protect a user’s list of ‘safe’, previously launched skills. Solutions to protecting cloud data include using blockchain technology, in which an immutable, time-stamped database ledger records assets across a network of computers (nodes) in a ‘block’ (Kahn & Salah, 2018). However, due to the significant computational power needed for various authentication processes between nodes, Dorri et al. (2019) suggests that, due to the lower computational power of IoT devices, the use of a lightweight blockchain technology is more appropriate, wherein authentication methods are altered so that they are not as computationally intensive. This lightweight blockchain technology would be utilised to protect a user’s ‘safe’ list of previously searched skills, together with the ‘launch words’ attached to each trusted skill, ready to launch without confirmation should there be a close match when a search is performed on the user’s uttered request (see figure 1).

Figure 1

How SecureRate and lightweight blockchain (Dorri et al., 2019) will be utilised within the AWS IoT platform, based on Pathak et al. (2022, p.2), Brinson, Winner & Brooker (2021, p.213)

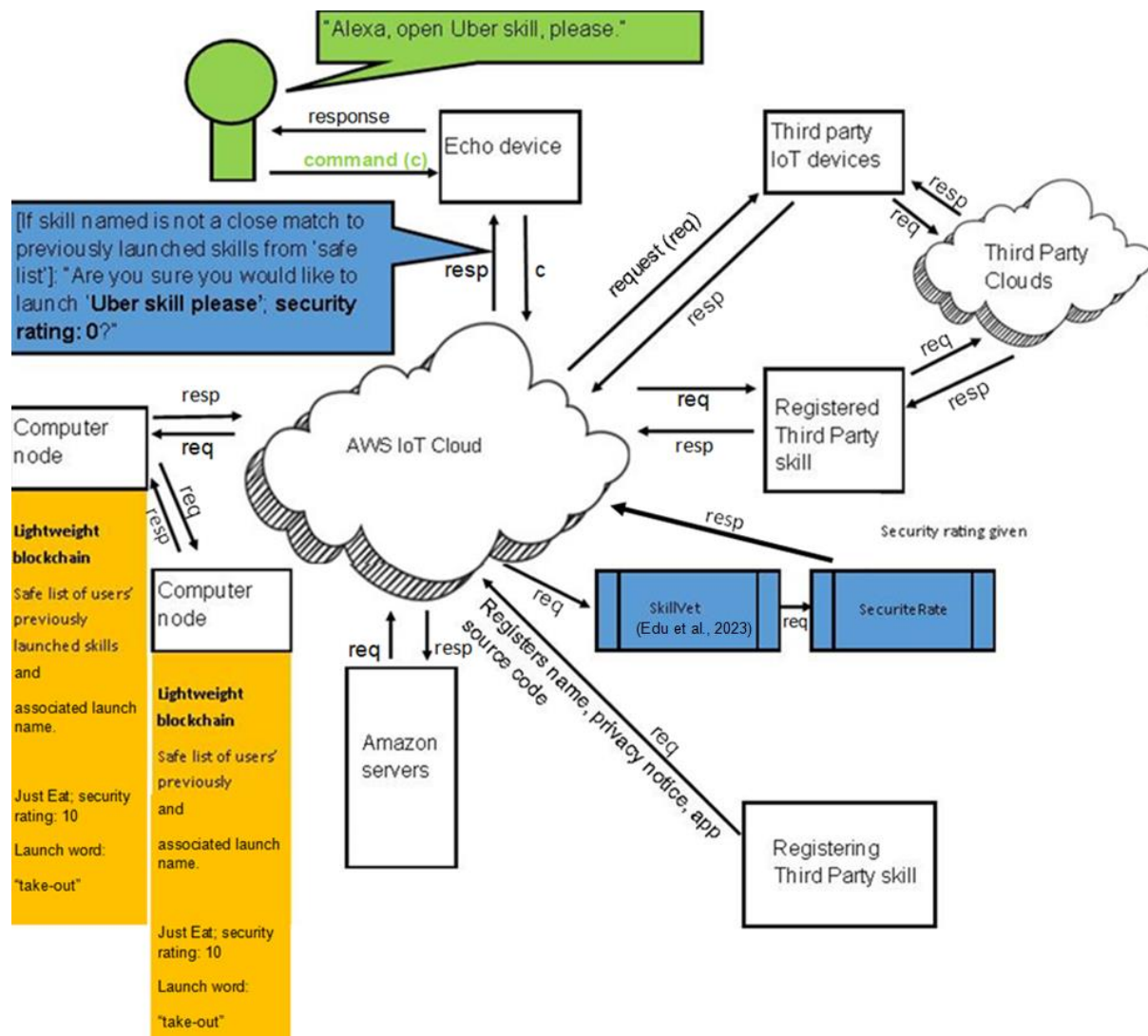
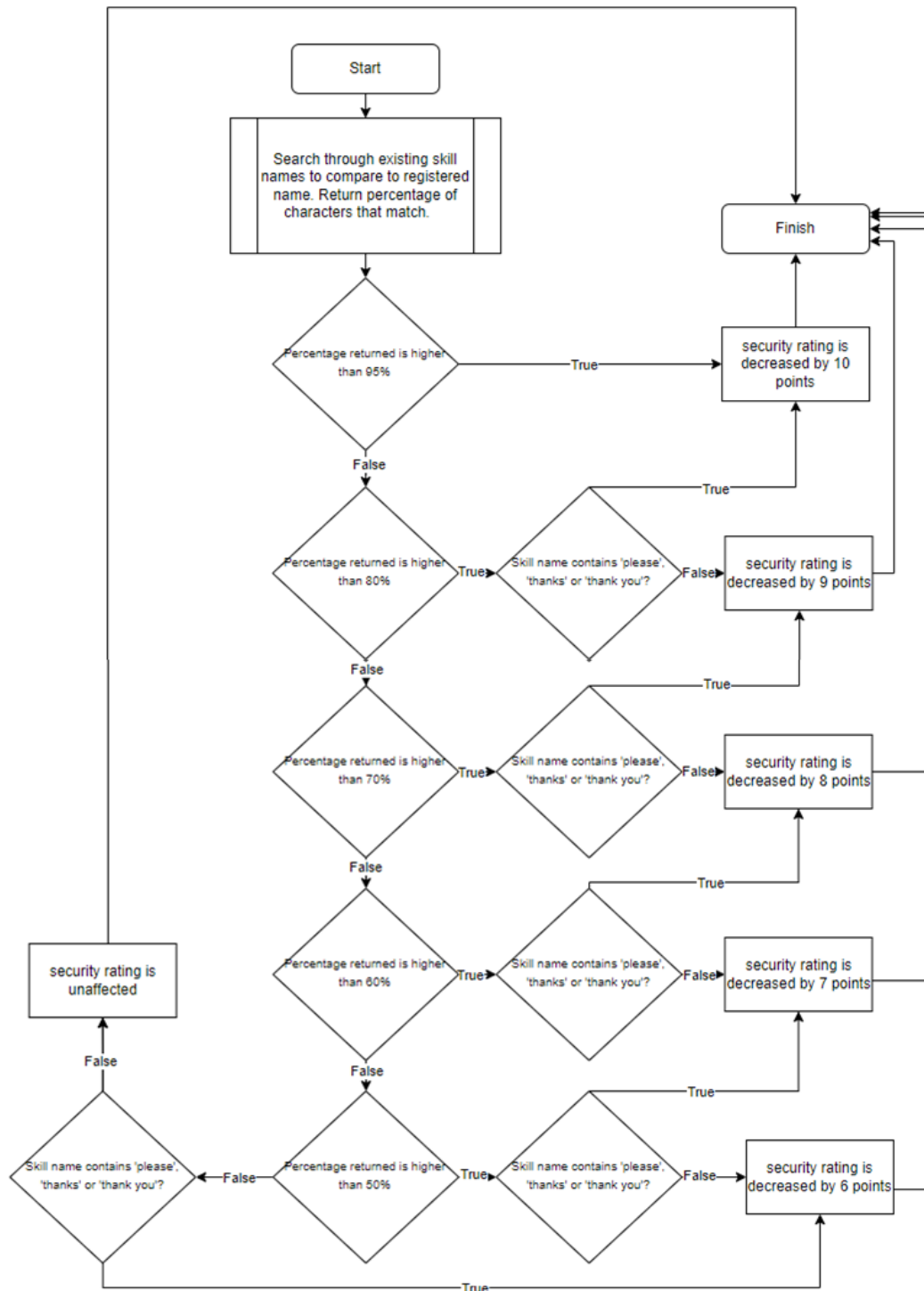


Figure 2

A flow diagram for a key algorithm within the SecureRate program:



Babun et al. (2021) suggests that the end user should choose their IoT platforms based on the out-of-the-box security it can offer, and not purely on how much functionality can be offered, due to the fact that increased functionality often equates to increased reliance on external computation and hence a wider attack surface. However, studies have found that loyalty towards a particular SPA – and therefore a choice of the IoT platform that it is a part of – is made on the grounds of the existing popularity of that device or on which device offers most functionality (Knote et al., 2021; Fernandes & Oliviera, 2021). It appears to be the case that, when considering investing in SPA devices and IoT interactivity, functionality takes precedence over security for the general public, or there is at least an ignorance of the existing security issues, perhaps due to the trust that users place on companies such as Amazon – trust that perhaps comes from such companies being market leaders. Offering a more secure service therefore could be seen to fix a problem that, in the eyes of most users, does not exist, and so Amazon would first have to admit that the problem does indeed exist; it is likely that such an admittance would risk driving down sales of Echo devices and risk discouraging continued use of existing devices, driving away developers and therefore investment in the AWS IoT platform. However, with the year-on-year increases of IoT devices, and the growing number of Amazon skills (Sinha, 2023; Edu et al., 2023), it seems likely that more and more security issues will arise and Amazon could risk associated negative press, which has been found to be a very strong motivating factor for tech companies to enforce change (Brookshear, J., & Brylow, D., 2020). Indeed, Lyon (2024) emphasises how cybersecurity knowledge within the general public will be extremely important as internet technology continues its rapid growth, particularly if we are to protect the most vulnerable members of society. It could therefore be prudent of Amazon to be

proactive rather than reactive, establishing itself as the most secure option for SPA and IoT integration in a landscape that is likely to be facing an increased level of security risk. This paper suggests that transparency on Amazon's part, and vigilance on the end user's part, made possible by the suggested solution – a security rating given as a result of implementing SkillVet (Edu et al, 2023) and SecureRate programs, combined with a confirmation message from Alexa before skills are launched - is needed to better safeguard users on Echo devices and make AWS IoT the secure choice for SPA users for decades to come.

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